# fm4npp-lartpc Research Summary

November 3, 2026 · Conversation research log

## Project Overview

The `fm4npp-lartpc` repository (primary upstream: HaiwangYu/fm4npp-lartpc, forked by JoshHenryPeterson) is a research codebase for a self-supervised **multi-modal foundation model** for the MicroBooNE Liquid Argon Time Projection Chamber (LArTPC). The model — called **qlfm** (charge + light foundation model) — jointly tokenizes TPC wire-plane charge hits and PMT scintillation light pulses, pretrained on 747k MicroBooNE open-data events using a masked autoencoder (MAE) objective. It is then evaluated on four downstream physics tasks with frozen or fine-tuned heads using small amounts of labeled data.

**Main result:** putting PMT light into the backbone during pretraining (*early fusion*) is substantially better than adding light afterwards via a cross-attention head (*late fusion*), even when the backbone architecture, objective, and compute budget are otherwise identical. This holds across all four downstream tasks and is most pronounced at low label fractions.

**0.999**
v/cosmic AUC (qlfm-M CL, 1% labels)

**0.892**
Hit F1 (qlfm-M CL, frozen 1%)

**0.713**
Flavor macro-F1 (qlfm-M CL, 1%)

**21M**

Parameters, best model (qlfm-M)

The project is part of the broader FM4NPP effort (Foundation Models for Nuclear and Particle Physics, arXiv:2508.14087, ICLR 2026), which introduced the first large-scale generalizable AI foundation model for nuclear and particle physics. The lartpc sub-project is a focused study of multi-modal fusion on MicroBooNE open data.

## Sources Read and How They Were Obtained

**Access limitations encountered.** GitHub blocks automated fetching ( `ROBOTS_DISALLOWED` ). The specific URL provided — `github.com/JoshHenryPeterson/fm4npp-lartpc` — could not be read directly. The upstream repo ( `HaiwangYu/fm4npp-lartpc` ) was accessible via its GitHub page. The Google Doc required sign-in and could not be fetched via the web tool. The two PDFs were eventually uploaded directly to the conversation and read in full.

| SOURCE | HOW OBTAINED | CONTENT |
|---|---|---|
| `HaiwangYu/fm4npp-lartpc` GitHub repo | Web fetch of the GitHub page (robots-compliant); search for PR/issue URLs | File structure, PR #2 (Round 1), PR #4 (Round 2), Issue #5 (Round 3 fairness controls) — all read in full |
| `survey-round1.pdf` | Uploaded directly to the conversation | Full 6-page survey written by Haiwang Yu, Nov 2026 — positioning, literature, and planned experiments E1–E5 |

| SOURCE | HOW OBTAINED | CONTENT |
|---|---|---|
| `Aobo_Li_Papers.pdf` (Google Doc export) | Uploaded directly to the conversation | List of 11 papers by or involving Aobo Li (UC San Diego) |
| Aobo Li papers (individual) | Web search + arXiv fetch for each paper listed in the Google Doc | Read in full: AP-SVM/LEGEND (arXiv:2410.14701), RESuM (arXiv:2410.03873), A-CNN/ XENONnT (arXiv:2603.23549), Density-guided CNP (arXiv: 2609.13593), DARWIN anomaly detection (arXiv: 2410.00755), KamNet (PRC 107, 014323) |
| Comparison FMs (PoLAr-MAE, Panda, FM4NPP) | Web search + arXiv fetch | Architectures, datasets, and downstream results from each paper read to understand the competitive landscape |
| BNL AI meeting agenda (Aug 2026) | Web search surfaced a mailing list post | Confirmed the survey PDF title: "Surveying Foundation Models for Scientific Research and Exploring Opportunities in LArTPC" |

## Project Progress: Rounds 1–3

### Round 1 — Prototype (merged Oct 16, 2026)

Built the full `qlfm` pipeline on MicroBooNE open data: HDF5 reader, GPU tokenizers (wire-time cell tokens for charge, per-PMT time-profile tokens for light), FlashAttention Transformer backbone, and multi-modal MAE with masked charge strips and masked light. Trained 6.6M ( `qlfm-S` ) and 21M ( `qlfm-M` )

models on up to 90% of the 747k-event dataset, evaluated on four downstream tasks:

- **T1** — hit-level ν/cosmic separation (F1)
- **T2** — event-level ν/cosmic separation (AUC)
- **T3** — neutrino flavor (νμCC / νeCC / NC) macro-F1
- **T4** — neutrino energy resolution

Key findings: pretraining helps most with few labels; light is decisive for T2 (event AUC 0.97 with light vs 0.65 without); the 21M model scales better than the 6.6M model; both outperform Pandora baselines on T1 and T2. Total compute: ~29 Polaris node-hours.

**Round 2 — Comparison with References (merged Oct 21, 2026)**

Benchmarked qlfm against four published foundation model recipes — FM4NPP (Mamba2), Panda (self-distillation), PoLAr-MAE (masked point modeling), and qlfm-SD (Panda-v2-inspired self-distillation) — plus Pandora direct and Pandora + BDT using features from the open-data reconstruction. All models received the same late-fusion cross-attention head ( `LateLight` ) for a common light interface.

| MODEL | EVENT AUC (T2) | FLAVOR F1 (T3) | ENERGY RES. (T4↓) | HIT F1 (T1) |
|---|---|---|---|---|
| **qlfm-MCL** | 0.999 | 0.713 | 0.284 | 0.892 |
| Panda (best other FM) | 0.978 | 0.679 | 0.343 | 0.715 |
| Pandora + BDT | 0.976 | 0.632 | 0.342 | — |
| PoLAr-MAE | ~0.95 | — | — | 0.537 |

| MODEL | EVENT AUC (T2) | FLAVOR F1 (T3) | ENERGY RES. (T4↓) | HIT F1 (T1) |
|---|---|---|---|---|
| FM4NPP (m3) | ~0.94 | — | — | 0.489 |

Compute: ~38.7 node-hours. 136-page tech note with 120 event displays.

**Round 3 — Fairness Controls (Oct 22–23, 2026)**

Issue #5 identified several confounds in the Round 2 comparison: the late-fusion head was weak (one pooling layer), FM4NPP had a 16,384-hit cap affecting 47% of T1 events, and qlfm had been iterated on this dataset while references got one untuned config. Round 3 ran four fairness controls:

1. **Strong late fusion** — 2-layer cross-attention head (256-d) for all backbones. Panda's hit F1 rose from 0.715 → 0.792; qlfm-M CL still led on all tasks at 1% labels; at 10% labels Panda matched qlfm on flavor and energy.
2. **qlfm-M C (21M, charge-only)** — without light in pretraining, the same model fell below Panda on all four tasks. Confirmed that qlfm's lead comes from early light fusion, not architecture.
3. **T1 hit-embedding fairness** — the per-hit embedding did not explain qlfm's T1 lead; a common raw adapter helped references slightly but left them well below qlfm.
4. **FM4NPP size/cap** — removing the hit cap added only +0.03 F1; m4 (19.5M) was at most marginally better than m3. FM4NPP and qlfm-M C form a second tier behind Panda.

Compute: 20.5 node-hours. The project now has a clean, defensible multi-modal claim with fairness controls documented.

## Planned Experiments (survey-round1.pdf, E1–E5)

The survey document, written by Haiwang Yu (Nov 2026, with Claude assistance), asks what is needed to publish at an ICML workshop. It identifies

five experiments totaling ~85–110 Polaris node-hours. Priority if time is short: **E2 and E5 first** (cheap, strengthen the mechanism story), then **E1** (the headline figure), then E3 and E4.

### E1 — Fusion-depth ladder at matched compute, with seeds

Light enters at input (early), after L/2 blocks (mid), or only at the head (late/XF), plus charge-only. All at qlfm-M size; also at S and ~60M. Two to three compute budgets each; frozen heads on T1–T5 at 1% and 10% labels with 3 seeds. Delivers a downstream-score-vs-compute curve per fusion depth.

~55–70 nh · Headline figure

Answers: does early fusion stay ahead as compute scales?

### E2 — Light-swap counterfactual and self-supervised flash matching

Replace each event's light by another event's (or shift it in time). Measure: (a) drop in T2 and T1 for early vs late fusion; (b) MAE cross-modal reconstruction error as a flash-matching score, retrieved against K candidates. Compare with Pandora's `flashmatch_score`.

~2–4 nh (inference only) · Cheap — high priority

Answers: is the early-fusion gain a physical charge–light consistency, not just flash presence?

### E3 — Missing and degraded modality robustness

Evaluate with light removed, 4/8/16 random dead PMTs, ±10–20% gain shifts and time shifts. Add one pretraining with modality dropout so a single model serves C and C+L.

~8–12 nh

Answers: does early fusion degrade gracefully? Relevant for running experiments where channels die.

**E4 — Time-aware light objective**

The hand scan showed the beam flash smeared by per-bin log-MSE. Use finer bins around the beam window and an earth-mover's distance (EMD) loss on the per-PMT time profile plus total PE. Report flash-time resolution and beam-window PE.

~8–12 nh (one pretraining + T5 heads)

Success criterion: flash time predicted to $\lesssim$0.5 µs, beam-window PE within 25%

**E5 — What does light teach the charge encoder?**

Linear probes on charge tokens only (light masked at inference) of CL vs C encoders for drift coordinate x, deposited energy, and v vertex. Layer-wise CKA between the two encoders. Attention maps charge→PMT for interpretability.

~2 nh Cheap — high priority

Success: CL encoder's charge tokens encode $x/t_0$ that the C encoder cannot

The survey also lists three optional low-cost additions: (a) release the MicroBooNE multi-modal benchmark (T1–T5 splits, tasks, baselines); (b) add a light→charge prompt (vertex z and charge centroid from light only); (c) ablate the per-modality reconstruction target (log-PE regression vs tokenized light).

## Additional Experiments Suggested by Aobo Li's Papers

The five experiments below are **not covered by E1–E5**. Each is derived from a specific methodological lesson in the Aobo Li papers listed in the Google Doc. All five are either inference-only checks costing under 1 node-hour each, or one additional pretraining (~3–5 nh), and fit within the remaining budget.

| # | EXPERIMENT | SOURCE PAPER | SURVEY GAP ADDRESSED | COST |
|---|---|---|---|---|
| **A1** | Data-MC consistency | A-CNN / XENONnT | Survey flags: "The v light is simulated; no beam-on | |

| # | EXPERIMENT | SOURCE PAPER | SURVEY GAP ADDRESSED | COST |
|---|---|---|---|---|
| | check: run charge→light cross-modal prediction on beam-off (cosmic-only) real data; compare PMT token distributions and T5 reconstruction error against simulation | (arXiv: 2603.23549) — Section IV.B on data-MC validation across three calibration datasets, with discrepancy reported as a systematic uncertainty | data." If the optical simulation's PMT profiles differ from real data, the entire early-fusion advantage could be a sim artifact. No experiment in E1–E5 touches this. | Inference only, ~0.5 nh |
| **A2** | Null-label shuffle test for T2 (v/ cosmic AUC); per-plane feature ablation for T1 and T2 — zero out each charge plane and each PMT group and measure the performance drop | "Beyond a Better Score" [A3] (Ma et al., arXiv: 2610.14024) — cited in the survey as a reason to present the light-cut audit as a validity check against score gaming | T2 AUC = 0.999 is an unusually high number on simulation-only data. Reviewers will ask whether it reflects genuine physics or an artifact of the overlay procedure (e.g., bookkeeping differences between simulated ν and cosmic events). E2 swaps light externally; this tests for internal shortcuts. | Inference only, ~0.5 nh |
| **A3** | T1 hit efficiency as a function of drift | Density-guided CNP (arXiv: 2609.13593, Ma & Li) — | The project reports a single global T1 F1. Drift-coordinate x requires $t_0$ from the light, so an | Inference only, ~0.5 nh |

| # | EXPERIMENT | SOURCE PAPER | SURVEY GAP ADDRESSED | COST |
|---|---|---|---|---|
| | coordinate x, hit amplitude, and wire plane — report efficiency curves (even simple histograms) showing CL vs C model uniformity in x | demonstrates that global hit F1 hides sharp local efficiency variations that matter for physics; proposes a data-efficient method to recover local efficiency curves with uncertainty bands | efficiency-vs-x curve directly and physically demonstrates what light teaches the charge encoder — more interpretably than a linear probe $R^2$ (which E5 uses). Zero extra cost beyond existing inference runs. | |
| **A4** | Pretraining-time augmentation of PMT tokens: random per-PMT gain scaling (±20%), time-offset jitter (±50 ns), and random zeroing of 1–2 PMTs during pretraining. Compare frozen-head performance against | A-CNN / XENONnT (arXiv: 2603.23549) — Section III.B, where targeted data augmentation (random S2 width upsampling, random noise injection) reduced data-MC discrepancy from ~10% to ~3% and made the model | E3 tests robustness at *inference* time. This tests whether the learned representations themselves are robust to known optical simulation discrepancies (photon visibility map errors, gain variations). Without this, a reviewer can argue the early-fusion advantage is fragile to calibration shifts. | ~3–5 nh (one retraining) |

| # | EXPERIMENT | SOURCE PAPER | SURVEY GAP ADDRESSED | COST |
|---|---|---|---|---|
| | standard pretrained model. | robust to known simulation-data differences | | |
| **A5** | Per-event charge→light reconstruction loss vs T2 ν/ cosmic score: plot these jointly for true-ν and true-cosmic events to check whether the MAE objective creates an internal shortcut (ν events having lower reconstruction loss and therefore easier T2 classification) | DARWIN anomaly detection (arXiv: 2410.00755) — trains a VAE on background only and uses reconstruction error as an anomaly score; validates that the score is not gamed by signal events that happen to be easy to reconstruct | E2 swaps the light signal externally. This tests for a subtler issue: whether the pretraining MAE loss itself creates a leakage pathway where ν events are intrinsically easier to reconstruct (because charge–light consistency is a simpler pattern to learn), making the downstream T2 head trivial. | Inference only, ~0.5 nh |

**Recommended order for A1–A5.** Do A2 and A5 first (cheapest, directly address the 0.999 AUC reviewer risk), then A3 (zero extra cost, strengthens E5), then A1 (addresses the sim-only data concern most critical for a physics venue), then A4 if budget allows. All five together cost at most ~6–7 node-hours beyond the E1–E5 budget.

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Sources: github.com/HaiwangYu/fm4npp-lartpc (PRs #2, #4; Issue #5) · survey-round1.pdf (H. Yu, Nov 2026) · arXiv:2603.23549 · arXiv:2609.13593 · arXiv:2410.00755 · arXiv:2410.03873 · arXiv:2410.14701 · arXiv:2502.02558 · arXiv:2512.01324 · arXiv:2508.14087. Literature checked Nov 3, 2026.